

CLASS 05 / SEP 22

MODULE 1 — FIELDS ATTRACTORS + SITE

DISTANCE DRIVES FORM

0.2 DUE

1.2 ISSUED TODAY / 1.1 DUE NEXT CLASS

AN ATTRACTOR IS A NUMBER

For every point: how far is it from something? Use that number to drive a property.

THE WHOLE IDEA

```
for each point p:  
    d = distance(p, attractor)  
    size = f(d)
```

That is it. Everything else
is a choice of f.

TWO NUMBERS, SAME JOB

INDEX told you where in the sequence.

DISTANCE tells you how far from something.

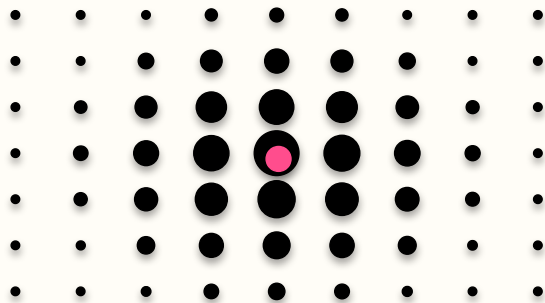
Both are just numbers driving a property — that is the only
idea in this module.

LAST WEEK YOU BUILT THE POSITIONS. THIS WEEK YOU MEASURE THEM.

THREE KINDS

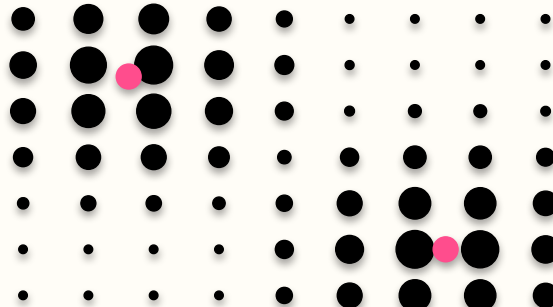
Same maths. What changes is what you measure the distance to.

SINGLE POINT



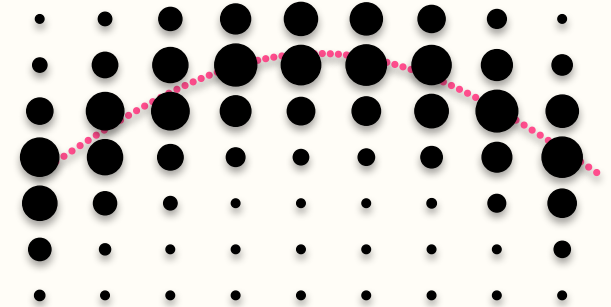
`distance(p, a)`

MULTIPLE POINTS



`min over all attractors`

CURVE



`distance to nearest curve point`

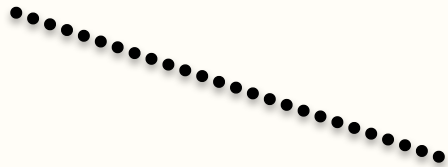
Whatever you measure to, the output is the same thing: one number per point. The attractor is not an object — it is a question you ask of every point.

FALLOFF IS THE FUNCTION

f is your decision. It is where the graphic quality actually comes from.

LINEAR

$$v = d$$

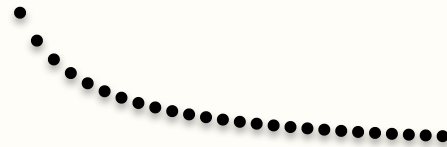


distance →

Even change all the way across.

INVERSE

$$v = 1 / d$$

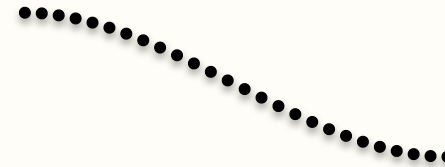


distance →

Sharp near the attractor, flat far off.

SMOOTH

eased



distance →

Soft at both ends. Usually the best-looking.

THRESHOLD

on / off at a cut



distance →

No gradient at all. Two populations.

Two students with the same attractor and different falloff produce completely different images. This is a design decision, not a setting.

THE FIELD IS AN IMAGE

Build it in 2D and look at it before you attach a single tile.

01 DISTANCE

Every point on the plane gets a number: how far to the attractor.

02 FALLOFF

That number is bent by f . Now it is a gradient you can see.

03 REMAP

The gradient is pushed into the range your property needs.

04 TILE

Only now does geometry appear. The field was already finished.

THE FIELD, READ AS A GREY VALUE ACROSS THE PLANE



IF THE GREY IMAGE IS NOT ALREADY A GOOD DRAWING, THE TILE WILL NOT SAVE IT.

1.2 requires one field submitted this way — on its own, as an image, with its falloff named.

REMAP

Distance comes out in model units. Your property needs a specific range. Remap is the bridge.

THE OPERATION

```
t    = (d - dmin) / (dmax - dmin)
out  = outmin + t × (outmax - outmin)
```

```
d    0 ... 8.4 units
out  0.1 ... 1.0 radius
```

SUBTRACT. DIVIDE. MULTIPLY. ADD.

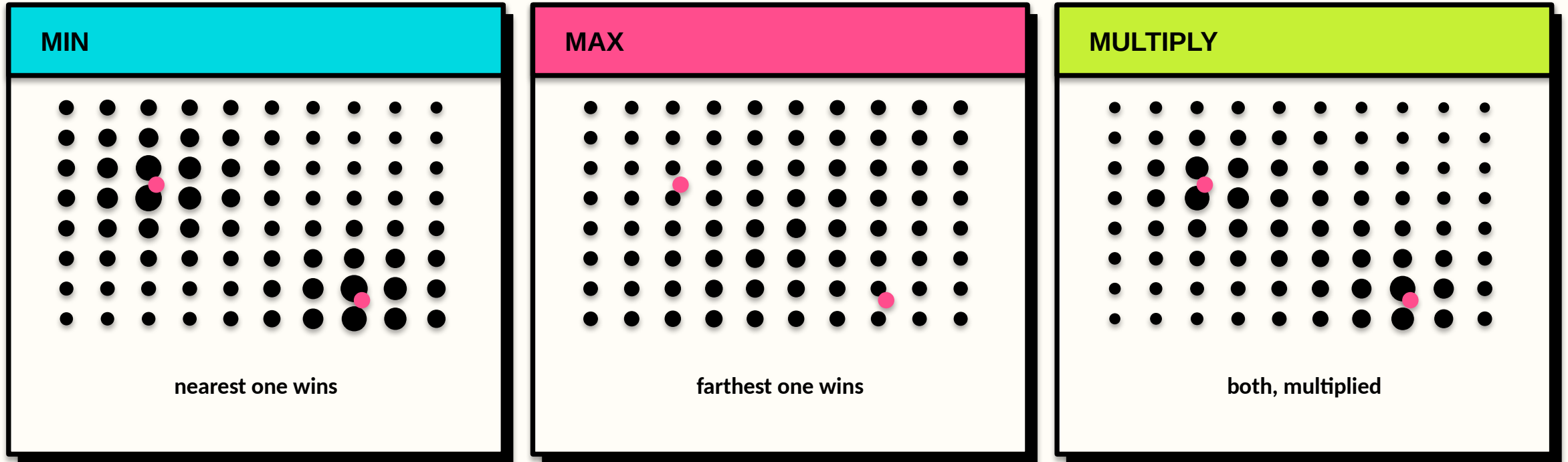
Nothing here is new. It is the arithmetic from Week 2, applied to a distance instead of a coordinate.

Get the ranges wrong and everything is the same size, or one element swallows the sheet.

1.2 REQUIRES YOUR REMAP RANGES DOCUMENTED ON THE SHEET — INPUT RANGE IN, OUTPUT RANGE OUT.

TWO ATTRACTORS, THREE ANSWERS

With more than one attractor each point has a list of distances. How you reduce it is a design choice.



Same two attractors in all three. Only the reduction changed — territories, a single central blob, or two peaks with a pinch between them.

UNDERNEATH, IT IS LISTS

Points, distances, sizes. Three lists, same length, same order.

POINTS	DISTANCES	SIZES	PARALLEL LISTS
[0] p_0	[0] 2.1	[0] 0.42	Item [2] in every list describes the same element. Break that correspondence — by sorting or culling one of them — and the sizes attach to the wrong points. That is what “managing lists” means, and it is next week's whole subject.
[1] p_1	[1] 0.4	[1] 0.95	
[2] p_2	[2] 5.8	[2] 0.10	
[3] p_3	[3] 3.2	[3] 0.28	
[4] p_4	[4] 1.0	[4] 0.71	

When a field looks scrambled, it is almost always a broken correspondence — not a broken formula.

THE SITE IS ALSO A LIST

A real place, imported, is a few hundred thousand points. You already know what to do with points.

01

PICK THE SITE

Somewhere you know. Google 3D Tiles, streamed into Blender.

02

IMPORT THE TILE

Photorealistic mesh, georeferenced. Set the origin and the scale before anything else.

03

SAMPLE TO POINTS

Distribute points on the surface, or take the vertices. Now it is a point cloud.

04

DECIMATE

Half a million points will not survive an instanced module. Cut it down and state both counts.

THE DIFFERENCE FROM LAST WEEK: YOU DID NOT PLACE THESE POINTS. YOU RECEIVED THEM.

A DIAGRAM MAKES A CLAIM

A render shows you the site. A diagram argues something about it. 1.2 wants the second one.

A PICTURE OF A SITE

Shows what is there.
Reads the same from any angle.
Could be of anywhere.

Nothing has been measured, so nothing can be wrong — and nothing can be argued with.

A DIAGRAM OF A SITE

Shows one thing, measured.
Suppresses everything else on purpose.
Could only be of this place.

Someone can look at it and disagree. That is the test.

THREE THINGS YOUR SHEET HAS TO STATE

WHAT YOU MEASURED

WHERE THE ATTRACTOR SITS, AND WHY

WHAT VARIES, AND OVER WHAT RANGE

WHAT COUNTS AS SITE INFORMATION

Your choice of subject. Not your choice whether to defend it.

ACCESS + MOVEMENT

Entrances, desire lines, stops, crossings, parking. Distance from how people actually arrive.

ENVIRONMENT

Sun and shade, prevailing wind, noise sources, canopy, water, elevation and flood line.

PROGRAM + USE

What each mass does. Density, frontage, vacancy, opening hours, where people stop rather than pass.

EDGE + BOUNDARY

Property lines, setbacks, thresholds, walls, the seam where one condition becomes another.

PICK ONE AND GO DEEP. FOUR SHALLOW LAYERS IS A MAP, NOT A DIAGRAM.

The attractor is where your claim lives. “The field intensifies toward the north entrance” is a claim. “I put a point there because it looked good” is not — and it is visible in the drawing either way.

WHEN TWO LISTS MEET

4 points, 2 attractors. How do they pair up? The software decides — unless you decide.

LONGEST LIST



POINTS



ATTRACTORS

4 results

The short list repeats its last item.

SHORTEST LIST



POINTS



ATTRACTORS

2 results

Stops when the short list runs out.

CROSS REFERENCE



POINTS



ATTRACTORS

8 results

Every point against every attractor.

This is why your two-attractor field went wrong. Longest List is the default, and the default is a decision someone else made for you.

GRAFT, FLATTEN, WRAP

Three operations that change list structure rather than list contents.

GRAFT

one list → one branch each

Each item becomes its own branch, so the next operation treats them separately instead of pairing them off.

FLATTEN

many branches → one list

The opposite. Useful at the end — destructive in the middle, because it throws away the structure you built.

WRAP

index past the end → back to the start

$i \bmod n$. An index of 6 in a list of 4 becomes 2. Cheap repetition, and the source of a lot of accidental patterns.

Structure is not content. Every one of these leaves the numbers alone and changes only who is grouped with whom.

ISSUED TODAY

1.2 — POINT: SITE ANALYSIS FIELDS

STUDY A REVIEWED CLASS 6 — SEP 29 · DUE CLASS 8 — OCT 13, WITH THE MIDTERM / 7%

An attractor turns distance into a value you can draw with. Study A builds that instrument on a flat plane, where you can see exactly what it does. Study B points it at a real site and uses it to diagram something you found there.

STUDY A

THE INSTRUMENT

Attractors on a flat field, driving your 1.1 tile. The field shown as an image before it drives anything.

STUDY B

THE SITE

A site of your choosing, imported as a point field, diagrammed with attractors anchored in what you found there.

STUDY A IS PINNED UP IN TWO WEEKS. IT IS NOT A DRAFT OF STUDY B — IT IS HOW YOU LEARN THE TOOL.

REQUIREMENTS

Counting these earns a C. The rest is judgement.

3	attractor types across the set: single point, multiple points, curve
1	field shown on its own as a 2D image, before it drives geometry, falloff named
3+	distinct parameters driven by distance — scale, rotation, density, height, colour
1+	explicit remap, with input and output ranges stated on the sheet
1	site imported and reduced to a point field — counts stated before and after, scale and origin declared
1	site condition measured and named — access, environment, use, or edge
1+	attractor whose position is argued from that condition, in one written sentence
2	sheets minimum, covering both studies, in your visual identity

Falloff built, not called from a preset. The imported site is expected — a preset attractor plugin is not.

NOW — 1.1 CLINIC, THEN YOUR FIRST FIELD

1.1 is due next class. Bring your problems to the front, then start Study A.

✓

Tile modeled, and three arrays running — which three, and why those

✓

One attractor working on a 1.1 array, any falloff

✓

The field looked at on its own, as a flat image, before the tile is attached

✓

Remap ranges written down, ready to put on the sheet

✓

A site named for Study B — and one sentence on what you want to find out about it

THREE DIFFERENT SCALES IS NOT THREE PARAMETERS.

BEFORE NEXT CLASS

FINISH

1.1 is due next class. Two sheets minimum, in your visual identity — precedent, tile, and three arrays.

INSTALL

The Google 3D Tiles importer in Blender, and get one tile of your site to load. Do this before the studio session, not during it.

WRITE

One sentence: what you want your site diagram to show. If you cannot write it, you do not have a diagram yet.

BRING

Study A running, even badly. Next class is a clinic and it is worth more with something on screen.